Negative Binomial models, intro to ZIP models

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Overdispersion

Overdispersion occurs when the response variable Y_i has greater variability than the model accounts for.

Poisson: $Var(Y_i|\mathbf{x}_i) = \mathbb{E}[Y_i|\mathbf{x}_i] = \mu_i$

quasi-Poisson models

$$\log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$
$$Var(Y_i|\mathbf{x}_i) = \phi\mu_i$$

Negative Binomial models

$$Y \sim NB(\mu, r) \qquad f(y; \mu, r) = \frac{\Gamma(y + r)}{\Gamma(y + 1)\Gamma(r)} \left(\frac{\mu}{\mu + r}\right)^y \left(1 - \frac{\mu}{\mu + r}\right)^r$$

$$Y_i | \mathbf{x}_i \sim NB(\mu_i, r) \qquad \log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Example: articles data

```
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.233528   0.117585   1.986   0.0470 *
phd          0.008203   0.036069   0.227   0.8201
ment         0.030374   0.003240   9.373   <2e-16 ***
kid5        -0.093256   0.046475  -2.007   0.0448 *

(Dispersion parameter for Negative Binomial(2.1647)
   family taken to be 1)

AIC: 3145.6
```

Inference for negative binomial models

- Can use the usual Wald tests for β (from asymptotic normality)
- LRT also works because a full distribution is specified

Example: articles data

```
m3 <- MASS::glm.nb(art ~ phd + ment + kid5, data = articles
m4 <- MASS::glm.nb(art ~ phd + kid5, data = articles)

pchisq(m3$twologlik - m4$twologlik, 1, lower.tail=F)
```

```
## [1] 1.708206e-18
```

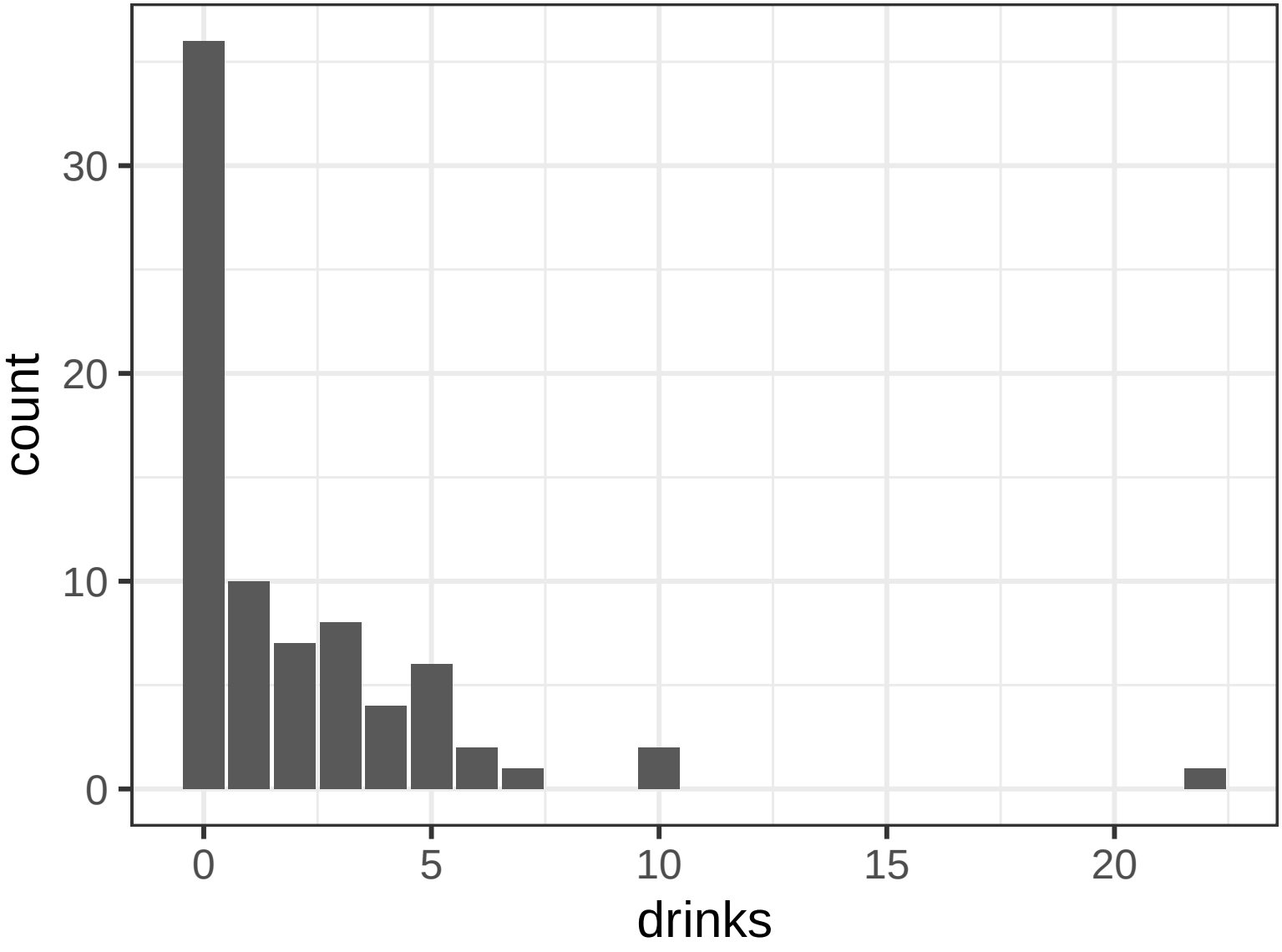
New data

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: number of drinks the student reports consuming
- `sex`: whether the student identifies as male
- `OffCampus`: whether the student lives off campus
- `FirstYear`: whether the student is a first-year student

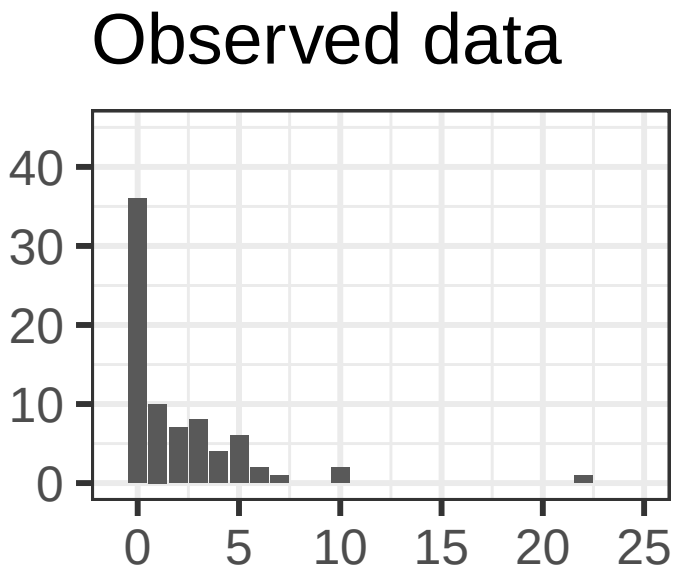
Our goal: model the number of drinks students report consuming.

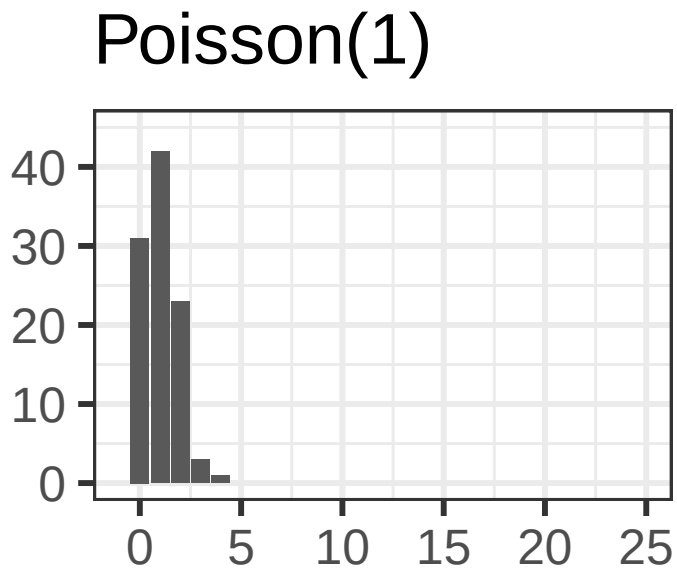
EDA: drinks

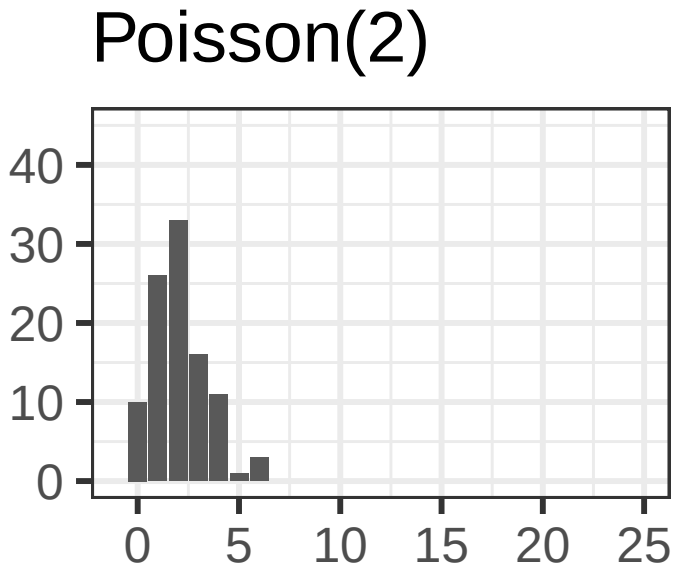


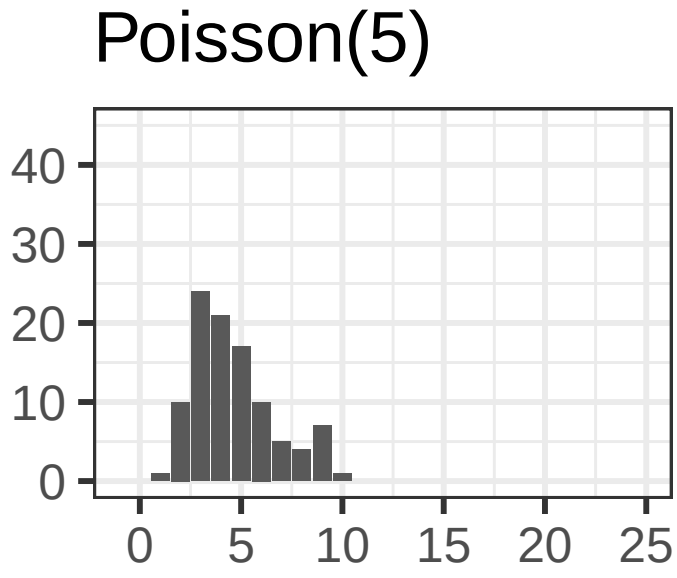
What do you notice about this distribution?

Comparisons with Poisson distributions



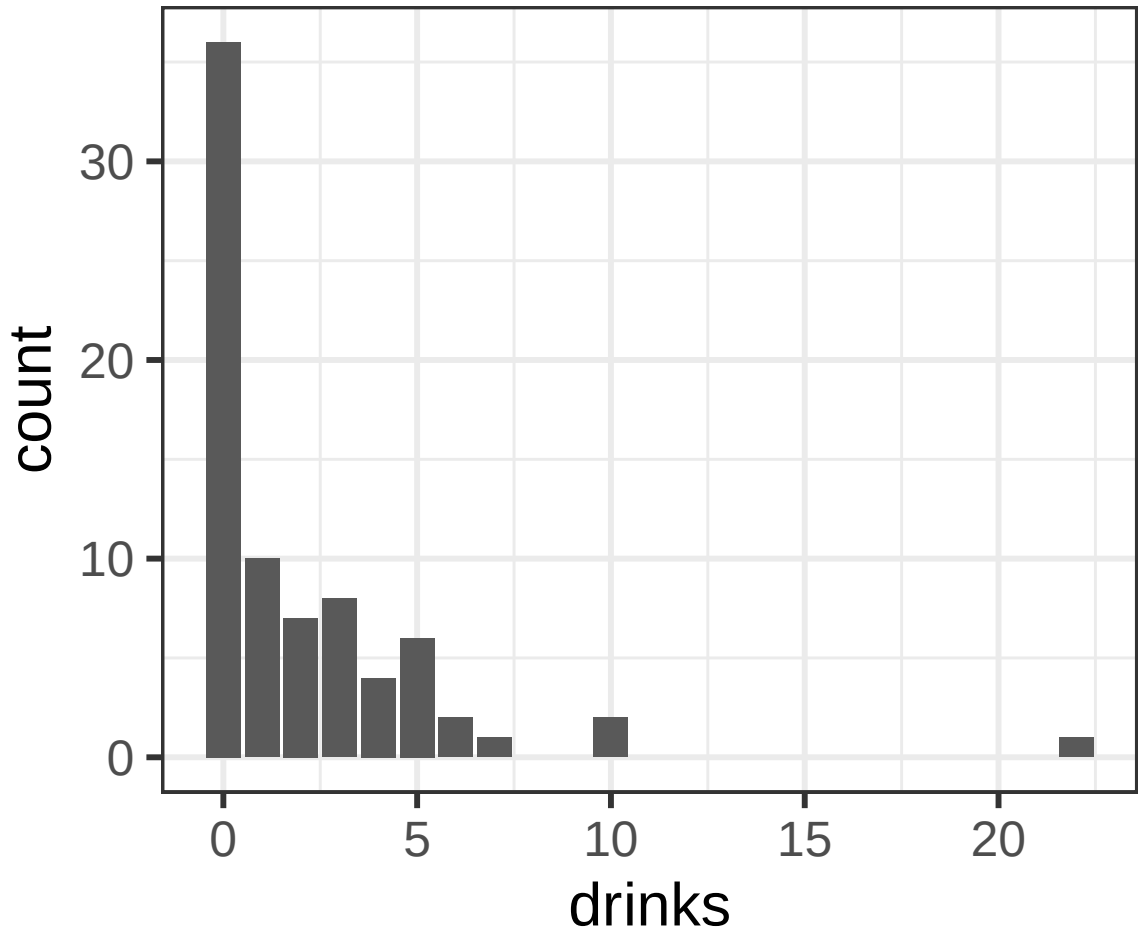






Excess zeros

Why might there be excess 0s in the data, and why is that a problem for modeling the number of drinks consumed?



Zero-inflated Poisson (ZIP) model

In R

```
library(pscl)

m1 <- zeroinfl(drinks ~ sex + FirstYear + OffCampus,
               dist = "poisson", data = wdrinks)

m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

Activity

Work on the activity (handout).

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated probability that a female first-year student who lives on campus never drinks?

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated average number of drinks for a second-year male student who lives off campus and sometimes drinks?

Activity

```
m1$coefficients
```

```
## $count
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##     0.8010435     0.9834693    -0.1619318     0.3723520
##
## $zero
##   (Intercept)          sexm FirstYearTRUE OffCampusTRUE
##   -0.39618515   -0.07079078    0.89197114   -1.69137146
```

What is the estimated probability that a male first-year student who lives off campus had at least one drink last weekend?